

OMID ZANGANEH

Senior GIS Developer and Analyst

MS Geography — Geographic Information Science & Technology · Workflow Automation · AI/ML Integration
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PROFESSIONAL SUMMARY

Senior GIS Developer and Analyst with a track record of turning manual spatial workflows into reliable production tools. Specializes in ArcGIS Pro SDK (.NET/C#/WPF), Python geoprocessing, SQL Server pipelines, and applied AI for engineering teams. Proven results: ~90% fewer manual GIS steps, multi-day workflows reduced to minutes, and AI-assisted contract sourcing in hours instead of months. 2025 Edison Award Nominee and 2026 Edison Award Winner at Olsson. MS Geography (GIS&T), University of Nebraska at Omaha, 4.0 GPA.

WORK EXPERIENCE

Olsson ♦ 2025 Nominee & 2026 Edison Award Winner

March 2025 – Present

GIS Developer and Analyst

Lincoln, Nebraska

- Worked directly with many telecom and infrastructure clients, delivering GIS tools, network design support, and spatial analysis that improved speed and consistency of delivery.
- Built production Python and C# GIS apps, including bore profile automation that cut processing from days to minutes.
- Created an ArcGIS Automation Suite of Python toolboxes that reduced manual GIS steps by ~90% and sped up fiber network design.
- Shipped ArcGIS Pro add-ins (.NET/C#/WPF) adopted across projects and teams — including a GIS Data Downloader used firm-wide, FTTH Network Designer, RF Analysis panel, and Street View tool.
- Learned iBwave to deliver DAS (distributed antenna system) designs, and built AI tools (Azure OpenAI, Google AI Studio, YOLO) that cut contract sourcing from months to hours.

University of Nebraska at Omaha

January 2024 – August 2025

Graduate Teaching Assistant — Instructor of Record

Omaha, Nebraska

- Taught Human-Environment Geography labs as sole instructor of record to 150+ students across three semesters.

University of Nebraska at Omaha

June 2024 – August 2025

GIS Technician — Omaha Spatial Justice Project

Omaha, Nebraska

- Digitized historical parcels and legal records into a geodatabase of racially restrictive covenants in Douglas County.

SELECTED PROJECTS

RFP Radar — AI-Powered RFP Sourcing

Azure OpenAI · Google Gemini · Playwright · Python/C#

Desktop app for web-grounded RFP search, classification, and CSV export. Months of sourcing → hours.

Bore Profile Automation — Directional Drilling Profile Generator

Python · C# · ArcGIS Pro · SQL Server · Matplotlib

Reads spatial waypoints and elevation models; generates 2D/3D bore profiles. Days → minutes.

Fiber Automatic Expansion — Density Grouping & Fiber Build-Area Planning

C# · .NET · ArcGIS Pro SDK · WPF · Spatial Analysis

Grids the study area for density analysis, flood-fills viable build zones by PPM thresholds, and generates economic scorecards. Used across multiple clients; days/weeks → minutes.

FTTH Network Designer — Automated Fiber Optic Network Planning

C# · .NET 8 · ArcGIS Pro SDK · WPF · Kruskal MST

Places shafts, connects homes, and builds an optimal Main Trunk / Terminal Branch backbone. Days → minutes.

ArcGIS Data Downloader Add-in — Multi-Source GIS Data Acquisition

C# · .NET 8 · ArcGIS Pro SDK · WPF · REST APIs

Downloads OSM, USGS, FEMA, Census/TIGER, Wikipedia, and BSL layers into ArcGIS Pro. Adopted firm-wide across projects and teams.

RF Analysis Tool — 8-Tool Wireless Planning Panel

C# · ArcGIS Pro SDK · Python

Coverage prediction, PCI/RSI planner, tilt/azimuth optimizers, interference analysis, and tower placement — all as map layers.

Aerial & Streetview AI Detection — YOLO Utility Infrastructure Detection

Python · YOLO · OpenCV · Aerial/Street Imagery APIs

Detects and classifies utility assets from aerial and street imagery; exports georeferenced results to ArcGIS.

GeoPipe — Enterprise GIS ETL to SQL Server

Python · CustomTkinter · pyodbc · SQL Server · PyInstaller

GUI ETL for large spatial/tabular imports with schema auto-detection, GEOMETRY/GEOGRAPHY support, and connection-loss auto-resume.

omidzanganeh.com — Full-Stack Portfolio & Browser GIS Tools

Next.js 16 · React · TypeScript · Leaflet · Vercel

Personal site with 10+ browser GIS tools (geocoder, isochrone, elevation, census, coordinate converter), Gym Flow, Stock Screener, and an AI news aggregator.

EDUCATION

Master of Science, Geography — Geographic Information Science & Technology December 2025

University of Nebraska at Omaha, Nebraska **GPA: 4.00** · **GRACA Project Award**

Thesis: Spatiotemporal Analysis of NOx Emissions from U.S. Cement Plants Using TROPOMI Data — remote sensing, hotspot analysis, and environmental justice.

Bachelor of Science, Geomatics (Surveying) Engineering

August 2016

Geomatics College of National Cartographic Center (GCNCC), Tehran

TECHNICAL SKILLS

Languages & Frameworks Python · C# / .NET · SQL · TypeScript · JavaScript · HTML/CSS · React · Next.js · WPF

GIS & Spatial ArcGIS Pro · ArcGIS Pro SDK · ArcPy · Python Toolboxes (.pyt) · ArcGIS Online · ArcGIS Enterprise · QGIS · Geoprocessing · Network Analysis · Remote Sensing · Google Earth Engine · iBwave

AI / ML Azure OpenAI · Azure AI Foundry · Google AI Studio / Gemini · YOLO · OpenCV · Prompt Engineering · Web Grounding · Batch Classification Pipelines

Data & Backend SQL Server · Supabase · PostgreSQL · pyodbc · ETL Pipelines · Spatial Types (GEOMETRY/GEOGRAPHY) · Smartsheet API

Web & Cloud Next.js App Router · Vite · Leaflet · Vercel · REST APIs · PWA · Microsoft Azure · Google Cloud

Desktop & Tooling CustomTkinter · PyInstaller · Playwright · Matplotlib · AutoCAD · ENVI · SNAP · Photomod · Tableau · Adobe Photoshop · Adobe Illustrator

Languages English (Fluent) · Persian (Native)

For project write-ups, live GIS tools, and demos: omidzanganeh.com
